

Individual or Teamwork of 2		
Student(s) identification		
Last Name	First Name	
Tariq	Muhammad Ibtisam	261265753

6 Points	Title of Incident:	2024 UnitedHealth Group Change Healthcare Ransomware Attack	
	Discovery date:	February 21, 2024, when UnitedHealth Group publicly disclosed that Change Healthcare had experienced a cybersecurity incident involving ransomware that disrupted national healthcare transaction systems.	
	Start date of Incident:	The initial compromise is believed to have occurred around February 12–14, 2024, when threat actors first gained unauthorized access to Change Healthcare’s network environment.	
	End date of incident:	Primary containment occurred in March 2024 after systems were isolated and rebuilt, though full restoration of services continued in phases into mid-2024 due to operational backlogs and recovery complexity.	
	Organization affected:	UnitedHealth Group, through its subsidiary Change Healthcare, a major healthcare technology and payment processing provider.	
		Number of employees	Approximately 440,000 employees worldwide.
		Revenues	Approximately \$371 billion USD in annual revenue.
		Country	United States, headquarters located in Minnetonka, Minnesota.
		Line of business	Healthcare insurance, pharmacy benefit management, and healthcare claims/payment clearinghouse processing.
4	Cause of the incident:	This incident resulted from a deliberate external ransomware attack. Threat actors gained unauthorized access to Change Healthcare’s systems using compromised credentials tied to a remote access portal. Reports indicated that multi-factor authentication was not enabled on the affected entry point, allowing attackers to establish persistence and move laterally within the network before deploying ransomware.	
	Aspect affected:	The primary principle affected was availability , as core healthcare transaction platforms were taken offline. Confidentiality was also impacted due to data exfiltration, and integrity risks existed because system encryption and unauthorized access could have altered transactional data.	
10	Incident description		
	In February 2024, Change Healthcare was targeted by the ALPHV/Black Cat ransomware group. After gaining network access, the attackers exfiltrated sensitive healthcare and financial data and deployed ransomware across critical systems. UnitedHealth disconnected affected infrastructure to contain the attack, which immediately disrupted nationwide claims processing, pharmacy adjudication, and provider payment functions. Because Change Healthcare operates one of the largest medical clearinghouses in the United		

	States, the shutdown extended beyond the organization itself and affected hospitals, clinics, insurers, and patients at a national scale.
10	Direct impacts
	The incident severely disrupted core healthcare operations. Claims processing systems were unavailable, preventing providers from submitting insurance bills. Pharmacy transaction platforms were also offline, delaying prescription approvals and forcing some patients to pay out of pocket. Provider reimbursement systems stopped functioning, creating financial strain across healthcare organizations. From a Business Impact Analysis perspective, these platforms represent mission-critical services with extremely low downtime tolerance. The lack of immediate failover capabilities amplified operational and financial consequences across the healthcare ecosystem.
5	Direct cost
	UnitedHealth reported billions in direct financial impact. Costs included ransomware payment, forensic investigations, infrastructure rebuilding, legal response, regulatory compliance, and emergency funding programs issued to healthcare providers unable to receive payments. Additional indirect costs stemmed from operational downtime and reputational damage.
10	Incident timeline
	Feb 12–14, 2024: Initial system compromise occurs Feb 21, 2024: Incident discovered; systems taken offline Late Feb 2024: Nationwide claims and pharmacy outages begin March 2024: Containment and early restoration phases start April 2024: Claims processing gradually resumes May–June 2024: Payment backlogs and breach response continue
5	Perpetrator
	The attack was attributed to the ALPHV/Black Cat ransomware group. Attribution confidence is high based on ransomware signatures, dark-web leak activity, and federal law enforcement reporting.
5	Motivation
	The primary motivation was financial extortion through ransomware deployment. Secondary motives included monetization of stolen healthcare data and leverage through double-extortion tactics. This assessment aligns with observed attacker behavior and ransomware group operating models.
10	Vulnerabilities being exploited and techniques used by the attackers
	The attackers exploited compromised credentials associated with a remote access system that lacked multi-factor authentication. After gaining entry, they conducted lateral movement, escalated privileges, exfiltrated sensitive data, and deployed ransomware payloads. Weak identity security controls and insufficient access segmentation enabled the attackers to reach systems supporting critical healthcare operations.
10	Incident resolution
	UnitedHealth responded by isolating affected networks, initiating forensic investigations, rebuilding compromised infrastructure, and restoring systems from backups. Service restoration occurred in phases, prioritizing pharmacy and payment platforms. Financial assistance programs were also launched to support impacted providers. While systems were eventually restored, stolen data and operational disruption could not be fully reversed.
5	Communications to external parties

	UnitedHealth issued public statements and regulatory disclosures throughout the incident. However, many healthcare providers reported limited visibility into restoration timelines. The evolving scope of the disruption created operational uncertainty for dependent organizations, highlighting the importance of structured crisis communication within continuity planning.
10	Long-term impacts
	The incident is expected to have lasting financial, regulatory, and reputational consequences. UnitedHealth faces legal exposure, government investigation, and increased cybersecurity oversight. The attack also exposed systemic risk within the healthcare sector, particularly the concentration of claims processing within a single clearinghouse provider. This dependency creates elevated operational risk if similar disruptions occur in the future.
10	Lessons learned
	This incident underscores the importance of strong identity security controls, including mandatory multi-factor authentication for remote access systems. It also highlights the need for mature Business Continuity and Disaster Recovery planning, particularly hot-site failover and alternate transaction processing capabilities. More robust dependency mapping and sector-wide contingency coordination could have reduced the scale and duration of the disruption. While UnitedHealth had response frameworks in place, gaps in preventive controls and recovery readiness contributed to the incident's severity.

REFERENCES

UnitedHealth Group. (2024). *Form 8-K: Cybersecurity incident disclosure*. <https://www.unitedhealthgroup.com>

UnitedHealth Group. (2024). *Change Healthcare cybersecurity incident update*. <https://www.unitedhealthgroup.com>

American Hospital Association. (2024). *Impact of the Change Healthcare cyberattack on U.S. healthcare providers*. <https://www.aha.org>

Cybersecurity and Infrastructure Security Agency (CISA). (2024). *Healthcare sector cyber threat advisory following Change Healthcare ransomware attack*. <https://www.cisa.gov>

U.S. Department of Health & Human Services (HHS). (2024). *Statement on national healthcare payment disruption following cyber incident*. <https://www.hhs.gov>

Reuters. (2024). *UnitedHealth unit Change Healthcare hit by ransomware attack*. <https://www.reuters.com>